

Executive Summary

We evaluated five public fraud datasets (IBM AML, IEEE-CIS, BankSim, Kaggle Credit Card, and Fraud Detection Handbook's synthetic data) against the requirements of an operational transaction fraud system. Each dataset has distinct strengths and weaknesses. The IBM **AML** dataset is a large-scale synthetic multi-bank transaction network (up to 180M transactions [97†L168-L172]) with explicit money-laundering labels and realistic laundering patterns [109†L181-L189] , but it lacks many real-world context features (e.g. merchant, device). The **IEEE-CIS** dataset is a real large-scale credit-card transaction set (590,540 transactions, ~3.5% fraud) [45†L1-L4] with rich anonymized identity features, but it is static and heavily obfuscated (features cannot be directly interpreted) [49†L488-L492] . **BankSim** is a small synthetic bank-transfer simulation (~594k transactions, 1.21% fraud) [39†L3-L6] with basic customer/merchant transactions; it has clear temporal sequences but few features and some unrealistic correlations [39†L3-L6] . The **Kaggle Credit Card** dataset (284,807 transactions, 0.17% fraud) [66†L1-L4] is real and widely used, but it contains only PCA-transformed features with no customer or merchant IDs, so no behavioral continuity or explainable context [120†L150-L158] [122†L163-L172] . The **Fraud Detection Handbook** synthetic dataset (~1.75–1.8M transactions, 5000 customers, 10000 terminals, ~0.8% fraud) [122†L199-L202] offers fully controlled behavior: it includes sequential transaction history, multiple fraud scenarios (threshold, compromised terminal, account takeover), and both categorical and numerical features [122†L193-L202] . Its code is open-source (GPLv3) [116†L130-L134] .

None of the datasets is a perfect plug-in for a live operational fraud workflow. The IBM AML data excels at *graph-based money-laundering patterns* but lacks merchant, device, and real customer demographics. IEEE-CIS and Kaggle credit-card data provide *real transaction examples* but only as isolated snapshots (no account histories or sequential context) and with heavy feature anonymization. BankSim and the Handbook data are synthetic: they provide temporal and behavioral structure but also include simplistic or unrealistic artifacts. Across all, we found various leakage risks (e.g. subtle correlations in BankSim, or simplistic rules in the Handbook data) that must be guarded against.

Key findings: IBM AML offers the most comprehensive *multi-bank* scenario and explicit laundering labels [109†L181-L189] , making it highly suitable for money-laundering use-cases. The handbook synthetic dataset is useful for developing end-to-end detection (it includes customer IDs and transaction sequences [122†L199-L202]) but is somewhat toy-like. The IEEE-CIS and Kaggle credit datasets have the advantage of being real, but their limited feature transparency and lack of continuity make them better suited for pure classification experiments than for operational decision support. Ultimately, we recommend **replacing PaySim** as the primary modeling dataset. For an operational fraud platform, the best strategy is to **use PaySim only for demonstration** and rely on a combination of richer datasets for modeling: e.g. use IBM AML (for AML scenarios) combined with an appropriate real credit-card dataset like IEEE-CIS for card-fraud scenarios. This approach maximizes data realism while avoiding the known pitfalls of PaySim-like synthetic leakage.

Dataset Evaluation Matrix

Dataset	Transactions (Fraud %)	Accounts/ Customers	Temporal Span	Feature Highlights	Realism & Behavior	Leakage Risk	Lice
IBM AML (Transactions)	up to 180M (~0.1%) 【97†L168- L172】	Synthetic multi- agency (unknown count, multi-bank IDs)	Long (simulated years)	Timestamp; from/ to bank & account IDs; amount; currency; <i>IsLaundering</i> label 【97†L168-L172】	Very high (multi-bank, multi- currency, laundering patterns)	Low (labels are generative; no known simulator leaks)	CDL Shar 【3† L248
IEEE-CIS Fraud	590,540 (3.5%) 【45†L1- L4】	~20,000 cards (card1–6 fields)	Weeks/ months (2020)	432 fields: transaction and identity data, including DeviceInfo, DeviceType, M (merchant category), D1–15, V1–339 (numeric)	Moderate (real CNP transactions, repeat by card)	Moderate (no obvious leakage, but heavy anonymization)	Kag (Cor Lice
BankSim	594,643 (1.21%) 【39†L3- L6】	1,500 customers (no separate account ID)	6 months (180 days)	Step (day); CustomerID; Age; Gender; zipOrigin; MerchantID; zipMerchant; Category; Amount; Fraud flag 【39†L3-L6】	Moderate (temporal customer sequences to merchants)	Low (simple simulation; known static correlations)	CC E Kag
Kaggle Credit Card	284,807 (0.17%) 【66†L1- L4】	(no IDs)	2 days (Sep 2013)	TransactionDT; Amount; V1–V28 (PCA); Class 【49†L488-L492】 【66†L1-L4】	Low (no continuity; anonymized features)	Low (anonymized PCA; possible trivial class imbalance leaks)	CC E (orig ULB Wor

Dataset	Transactions (Fraud %)	Accounts/ Customers	Temporal Span	Feature Highlights	Realism & Behavior	Leakage Risk	Lice
FD Handbook Synthetic	~1.75–1.8M (0.7–0.8%) 【122†L199- L202】 【87†L67- L69】	5,000 customers	183 days (Apr–Sep 2018)	TRANSACTION_ID, TX_DATETIME, CUSTOMER_ID, TERMINAL_ID, TX_AMOUNT; plus engineered features (e.g. counts) 【122†L199- L202】	High (customers & terminals repeat; explicit fraud scenarios)	Moderate (fraud scenarios are rule-based)	GPL (cod BY-S 【11 L13

Individual Dataset Reviews

IBM AML Dataset (Synthetic Transactions for Anti-Money Laundering)

- **Origin & Purpose:** The IBM AML dataset is a *synthetic* transaction generator from IBM Research (Altman et al., 2023) 【97†L168-L172】. It simulates a virtual economy of banks, companies, and individuals and explicitly labels laundering transactions 【109†L181-L189】. It was released publicly (via Kaggle) for research on AML and fraud detection.
- **Size & Structure:** Datasets (HI/LI small/medium/large) range up to 180 million transactions 【97†L168-L172】. Each record includes timestamp, source/target bank and account IDs, amount, currency, and a laundering flag. The **HI-Large** dataset alone has $\sim 1.8 \times 10^8$ transactions 【97†L168-L172】. Accounts are represented by UUIDs (hex strings), and multiple banks (numeric codes) allow multi-institution flows. The fraud rate is very low ($\sim 0.1\%$) 【97†L168-L172】. Data is in CSV format; no explicit customer profile features (only account IDs).
- **Behavioral Realism:** This is the richest dataset for *longitudinal and relational behavior*. It supports repeated accounts and customers across banks, multi-step circular flows, and drift. The underlying simulation explicitly models realistic money-laundering cycles (placement, layering, integration) 【95†L301-L310】 【97†L168-L172】. Temporal sequences are realistic (transactions over simulated time), and entity graphs are complex 【95†L333-L341】. It allows building graphs of account relationships.
- **Potential Leakage:** As a synthetic generator with ground-truth labels, it was designed to avoid built-in leakage. There are no “post-event” fields like balances or engineered features beyond the basic transaction fields. (In contrast, the older PaySim dataset had balance fields that leaked class.) We found **no documented leakage issues** in IBM AML. However, any modeling must guard against trivial patterns (e.g. repeated high-frequency transactions) as in any imbalanced data.
- **Operational Suitability:** While excellent for modeling money-laundering, it lacks many operational data fields. There is no merchant data, device info, or geo-location. Thus it is suited for *alert generation* (label = laundering) but not for building analyst evidence like merchant names or transaction receipts. Explanation would focus on graph patterns (e.g. “account A and B were involved in a cycle”), but human-readable context is limited.

- **Feature Richness: Only transaction-level features:** timestamp, amount, currency, source/target bank and account IDs. No customer demographics or external metadata. This simplicity avoids many potential leakages, but limits building side-information.
- **Fraud Realism:** The fraud scenarios (structured by the AML generator) closely resemble money-laundering (circular transfers, layering). It even includes multi-currency flows and some “natural” laundering via normal-looking transactions [95†L329-L337] . However, it does not model other fraud types like account takeovers or social engineering. Thus it’s ideal for AML modeling but not for e.g. point-of-sale credit fraud or mule account networks.
- **Limitations:** It is fully synthetic, so distributional biases exist. Some patterns (e.g. no merchant or location features) mean it may not trigger as a human analyst’s alert. Also, being new (2023), community usage is still small. Its documentation (Kaggle page, IBM GitHub) is moderate; the methodology is in a research paper [97†L168-L172] . The license (Community Data License CDLA-Sharing 1.0) allows use in research [3†L240-L248] .
- **Usage in Research:** Cited by recent work (e.g. Altman et al. 2023 [97†L168-L172] , Tworek et al. 2026 [109†L181-L189]) as a benchmark for graph/AML models. Because of its scale and detail, it is promising for state-of-art GNN studies, though few outside IBM have yet used it. Best practices include using graph algorithms to capture circular flows. Known pitfalls: synthetic patterns may not cover rare fraud variants.

IEEE-CIS Fraud Detection Dataset

- **Origin & Purpose:** This dataset was introduced via a 2019 Kaggle competition (organized by IEEE and CIS) for card fraud classification. It contains real card-not-present transaction data from an online payments processor. The intent was algorithm benchmarking. The data is anonymized for privacy.
- **Size & Structure:** ~590,540 transactions (590k) with ~3.5% fraud [45†L1-L4] . There are **no explicit “account” fields** beyond multiple card ID features (card1–card6), product category, and address information. Transactions include a timestamp offset (TransactionDT) but no absolute date. The label is `isFraud`. There are 432 columns: 394 numeric “V” features (PCA components), and 38 transaction/identity features (card, addr, device, etc).
- **Behavioral Realism:** Moderate. Real transactions include repeated card usage (e.g. same card_id appears multiple times), so some customer continuity is present. However, there is no unique customer identifier linking transactions beyond card features, and TransactionDT is anonymized as seconds from start, so building true customer histories is hard. There is temporal sequence (TransactionDT as time), but it is dataset-specific, making behavioral modeling limited. No notion of beneficiary or account relationships beyond what the card network implies.
- **Potential Leakage:** The data has already been heavily anonymized (PCA features). The known Kaggle tasks explicitly warned about “test leakage” (train/test splits had time-sorted splits). In general, the dataset has no obvious synthetic leakage (no balance fields or post-transaction variables). However, the many features mean overfitting to quirks is possible. Practitioners should use time-based splitting to avoid future information leakage.
- **Operational Suitability:** This is essentially a *flat classification dataset*. It does not support an investigative workflow. There are no transaction narratives, no merchant names (only coarse M1–M9 codes), and highly opaque features. The only explanation possible is statistical (“this transaction was flagged by a model on hidden factors”), not narrative evidence. So it is well-suited for training ML models, but poorly suited for human case review or alert generation.

- **Feature Richness:** Very high feature count (431 features) but **low interpretability**. It includes some useful fields (TransactionAmt, ProductCD, card and address indicators, device type/info, sale in attempt), but the majority (V1–V339) are uninterpretable. No raw customer attributes (age, income, etc) or contextual data.
- **Fraud Realism:** The fraud cases are actual (labeled), but we do not know the fraud type (likely stolen-card online purchases). It can model **card-present vs card-not-present fraud** scenarios, but no details on social engineering, layering, etc. It reflects modern e-commerce fraud in one dimension.
- **Limitations:** The heavy anonymization and lack of customer context means features cannot be easily interpreted or explained. The dataset's short span (few months) and unknown demographics limit real-world representativeness [120†L150-L158] [120†L159-L162] . Known critiques note that it lacks the six basic transaction attributes (ID, datetime, customer, terminal, amount, fraud) in raw form [122†L163-L172] .
- **Usage in Research:** Extremely popular for benchmarking supervised learning and deep networks (imbalanced classification). Many Kaggle solutions and papers exist. Best practices involve careful handling of missing data and feature selection. Pitfalls include ignoring data split leakage or treating PCA features as opaque.

BankSim Dataset

- **Origin & Purpose:** BankSim is a synthetic transaction dataset generated by Lopez-Rojas et al. (2016) to simulate a simple banking system for fraud research. It's available on Kaggle (user *ealaxi*) and intended for testing ML models. The simulation was tuned on Spanish banking data.
- **Size & Structure:** 594,643 transactions over 180 days [39†L3-L6] . There are 1,500 unique **CustomerID** values (simulated users) and 1,500 merchant categories. Fields: time step (day), CustomerID, Age, Gender, zipOrigin, MerchantID, zipMerchant, Category, Amount, and Fraud label [39†L3-L6] . There is *no* separate account ID; each transaction is a payment from one customer to a merchant. The fraud rate is $7,200/594,643 \approx 1.21\%$ [39†L3-L6] .
- **Behavioral Realism:** The data contains repeated customer transactions and time order (Step = day). Hence one can model customer spending over 6 months. However, no actual multi-hop fund transfers exist (only customer→merchant). The relationships are bipartite (customers vs merchants). No account-to-account flows, no concept of credit/debit accounts. The transaction frequency per customer is realistic (Poisson process), but underlying patterns are simplistic.
- **Potential Leakage:** Being synthetic and fairly small, it has no engineered leakage fields like balances. However, prior analysis has noted suspicious correlations – e.g. the “Category” field was found strongly *anti-correlated* with fraud and “Merchant” positively correlated [39†L3-L6] . That is, certain categories almost never have fraud. This artificial correlation means a classifier could cheat by category, reducing realism. Also the dataset has class imbalance by category built-in. These are forms of potential “learnable artifacts” that could be considered data leakage risk if not intended.
- **Operational Suitability:** This is a simple transaction list. It supports basic fraud detection and some investigation: one could trace a customer's transactions or see how often they pay certain merchants. But it still lacks many real-world fields (no device/location, no merchant names beyond ID, no KYC data). It is primarily a classification dataset. It does allow assembly of evidence at the transaction level (you see the customer, merchant, amount, etc), but that evidence is minimal.
- **Feature Richness:** Very limited. Only demographic proxies (age, gender, zip origin) and merchant info (category, zip). Good that it includes merchant category which PaySim lacked. No card, no device, no channel info. Timestamps are only a coarse day index.

- **Fraud Realism:** Fraud is modeled as stolen-card or account takeover type: random customers are flagged to have random high-value fraudulent purchases at merchants. This resembles one type of “card fraud” (mule scheme). It does *not* cover money laundering (no layering possible) or other schemes. Velocity may appear (e.g. few fraudulent transactions per fraudster). Realistic merchant categories allow testing of merchant-based features.
- **Limitations:** Quite small (hundreds of thousands of records) and relatively dated. Its generation logic is fixed and known; the unrealistic correlation of “Category” with fraud is a major criticism [39†L3-L6] . The dataset is 7 years old and not frequently updated. License is permissive CC. Many academic examples use it, but it has limited documentation beyond the original paper [39†L3-L6] .

Kaggle Credit Card Fraud Detection (ULB/Worldline)

- **Origin & Purpose:** Released by the ULB Machine Learning Group (Brussels) and Worldline, this real dataset was used in an academic challenge (ICDM 2015 and Kaggle 2013) for class-imbalance modeling. It contains anonymized transactions of European credit cardholders.
- **Size & Structure:** 284,807 transactions, with 492 frauds ($\approx 0.17\%$) [66†L1-L4] , covering 2 days of real card usage. Each record has 30 columns: `Time` , `V1` – `V28` (PCA components of original features), `Amount` , and `Class` . There are *no* identifiers for customers, cards, or merchants – it’s a flattened list. The `Time` is seconds elapsed (offset), not a real date. All feature names V1-V28 are transformed (unsuitable for direct interpretation).
- **Behavioral Realism:** Essentially none. Since there are no account or card IDs, one cannot group transactions by card or user. Every transaction is independent. The two-day window is very short. Thus, there is no customer history, no drift, and no multi-step context. It cannot support any longitudinal modeling – it is purely a static snapshot.
- **Potential Leakage:** It is already anonymized; no engineered features beyond PCA. The only risk is the extreme class imbalance or any trivial splits (e.g. if data is not shuffled properly). Its simplicity avoids typical simulator leaks, but also hides all context. There are no fields like balance, so target leakage in that sense does not occur.
- **Operational Suitability:** Minimal. It is purely for ML benchmarking. Analysts cannot derive evidence from it (no merchant, location, or customer info). No explanation can be provided other than “the model found patterns in the PCA space.” It is purely a classification dataset, not an investigation dataset.
- **Feature Richness:** Very low in raw terms. Only the transaction amount and anonymized components. There are no categorical or contextual variables. One cannot compute anything like “transactions per merchant” or “customer spending history.”
- **Fraud Realism:** The fraud cases are actual credit-card fraud events from the bank, so they are real scenarios. However, with only PCA features, any notion of fraud type is lost. This dataset does *not* include any organized fraud schemes (no account takeover, no structured attacks) – it just reflects a tiny sample of labeled fraud. It is used for anomaly detection exercises, not scenario modeling.
- **Limitations:** As noted by its creators, it has “obfuscated features and a limited time span” [120†L150-L158] [120†L159-L162] . Many practitioners point out that it lacks the six basic fields of a transaction (ID, datetime, customer, etc) [122†L163-L172] . It has served well as a standardized benchmark, but its representativeness is poor.
- **Usage:** Extremely common in fraud ML literature and Kaggle tutorials. Best used as a testbed for supervised anomaly detection techniques. Pitfalls include over-optimistic performance reporting (because the short span may not capture concept drift) and lack of interpretability.

Fraud Detection Handbook Synthetic Dataset

- **Origin & Purpose:** Developed by researchers at ULB/Worldline as part of an educational *Reproducible Machine Learning* book [122†L215-L222]. It's a **custom simulator** for credit-card-style transactions, released with open-source code (GPLv3) and a CC license [116†L130-L134] [122†L215-L222]. It was designed to fill gaps in real data by allowing controlled experiments.
- **Size & Structure:** The baseline synthetic dataset contains ~1.75–1.8 million transactions [122†L199-L202] over 183 days (April–September 2018) with 5000 customers and 10000 merchant terminals [122†L199-L202]. It includes the six canonical fields: `TRANSACTION_ID`, `TX_DATETIME`, `CUSTOMER_ID`, `TERMINAL_ID`, `TX_AMOUNT`, and `TX_FRAUD` [122†L163-L172]. The fraud rate is ~0.7–0.8% [87†L67-L69] [122†L261-L264]. There are also engineered features (e.g. counts per customer) after preprocessing. The code and raw data are fully available on GitHub [122†L215-L222].
- **Behavioral Realism:** This dataset excels in supporting behavioral modeling. Each customer and terminal repeats over time; transactions are timestamped, enabling true time series. The simulator deliberately inserts time-dependent fraud patterns (e.g. a small fraction of each customer's transactions are scaled, phased terminal compromise, fixed-amount threshold fraud) [84†L39-L47] [87†L21-L29]. Thus it includes account takeover scenarios and velocity (multiple large withdrawals), albeit via simplistic rules. Categorical features have high cardinality (many terminals), as in real life [116†L83-L90] [122†L199-L202]. The authors emphasize it captures "non-trivial relationships between features, and time-dependent fraud scenarios" [122†L195-L202].
- **Potential Leakage:** Because the fraud labels are added to generated transactions post-hoc, classic leakage is low. However, the fraud patterns are rule-based and known – e.g. *all transactions above \$220 are fraudulent* in one scenario. That exact threshold is clearly detectable by any model, so it's an artificial leakage of domain knowledge. Analysts must be aware that the model could easily learn the scenario rules (e.g. terminal or amount rules). Also, the class imbalance (0.7%) is known, though realistic. In summary, engineered "loopholes" exist by design, so usage should treat them as part of evaluation rather than hidden leakage.
- **Operational Suitability:** As an end-to-end simulator, it is *close to supporting* an analyst workflow. Raw data includes `CUSTOMER_ID` and `TERMINAL_ID`, so one could look up a customer's history or terminal's fraud rate. It lacks real-world metadata (no merchant names, no device ID, etc), but it does allow assembling evidence (fraud at an unusual terminal). The engineered features (post-preprocessing) help in modeling. Yet, explanations would still be formal (e.g. "customer X had unexpected high spend pattern"), as it's synthetic.
- **Feature Richness:** The raw fields cover core transaction info. After preprocessing (binary encoding, aggregations), there are 15 input variables [122†L203-L211]. These include time-of-day features (seconds, days) and aggregated features per customer/terminal. This is richer in terms of enabling standard ML models (they include everything needed). However, it has no personal attributes (age, gender) and no external cues.
- **Fraud Realism:** The scenarios approximate real fraud: fixed-threshold fraud is like single-victim spending, compromised terminal is like physical skimmers, and compromised customer is like stolen credentials. These cover *some* modern fraud (especially credential theft). It doesn't simulate money laundering or mule networks. The distribution of amounts, and the timing (some weekly patterns) are realistic-ish. In practice, it reveals issues practitioners face (imbalance, feature engineering needs) [122†L195-L202].
- **Limitations:** It is still a toy model. The threshold scenario (A above 220 always fraud) is simplistic. The number of compromised entities (3 terminals, 3 customers per day) is arbitrary. It may not

capture edge cases of real financial crime. Also, because it is openly known, any model can “cheat” by directly inferring the rules. Documentation is thorough (published in the JupyterBook and source code) [122†L215-L222]. Adoption is mainly academic (course/book users), not a benchmark community.

Comparative Analysis

Criterion	IBM AML	IEEE-CIS Fraud	BankSim	Kaggle Credit	FD Handbook Synth.	Ranks
Behavioral Realism	High (multi-bank flows, true labels)	Medium (real CNP, repeated cards)	Medium (temporal, repeated customers)	Low (no continuity)	High (explicit sequences)	1: IBM AML, FD HB synth; 3: IEEE, BankSim; 5: Kaggle CC
Leakage Risk	Low	Low-Med	Medium (correlated fields)	Low	Medium (engineered rules)	1: IBM, IEEE, CC; 3: BankSim, FDHB
Feature Richness	Medium (limited context)	High (431 fields)	Low (basic demographics)	Low (only PCA+Amount)	Medium (core + engineered)	1: IEEE; 2: IBM; 3: FDHB; 4: BankSim, Kaggle
Operational Realism	Medium (AML focus only)	Low (flat data)	Low (simple payments)	Low (just PCA)	Medium (has IDs & sequences)	1: FDHB; 2: IBM; 3: BankSim; 4: IEEE; 5: CC
Explainability	Medium (graph patterns)	Low (opaque PCA)	Low (simple data)	Low	Medium (explicit features)	1: FDHB; 2: IBM; 3: others
ML Suitability	High (massive, labeled)	High (balanced features)	Medium (smaller)	High (balanced challenge)	High (balanced, engineered)	1: IBM, IEEE, CC, FDHB; 5: BankSim

Criterion	IBM AML	IEEE-CIS Fraud	BankSim	Kaggle Credit	FD Handbook Synth.	Ranks
Decision-Support Fit	Low-Med (alerts only)	Low (no context)	Low (no explanation)	Low	Medium (IDs present)	1: FDHB; 2: IBM; 3: others
Preprocessing Effort	Medium (large scale)	High (many missing, PCs)	Low (few fields)	Medium (just scaling)	Medium (requires feature creation)	1: BankSim; 2: IBM, CC, FDHB; 5: IEEE
Documentation/Quality	Medium (paper + GitHub)	Medium (Kaggle docs)	Low (limited)	Medium (paper + Kaggle)	High (book + code)	1: FDHB; 2: IEEE, CC, IBM; 5: BankSim
License/Use	Open (CDLA)	Kaggle terms	CC BY	CC BY	Open (GPL/CC)	1: IBM, CC, FDHB, BankSim; 5: (tie) IEEE-CIS (Kaggle license)
Community Adoption	Low (new)	High (Kaggle)	Medium (some use)	Very High	Low (academic)	1: Kaggle CC; 2: IEEE; 3: BankSim; 4: IBM; 5: FDHB

Key: Best rank (1) down to worst (5) in each criterion. Overall, **IBM AML** and the **FD Handbook synthetic** data score highest on behavioral realism and ML suitability, though with different focuses (AML vs credit). IEEE-CIS and Kaggle Credit are real and widely used, but score low on operational realism. BankSim is modestly realistic but limited in features. Licensing and community adoption favor the Kaggle datasets, while documentation and transparency favor the Handbook dataset.

Leakage Risk Assessment

- **IBM AML:** No obvious leakage. All features are pre-transaction fields (timestamp, IDs, amount, currency). The *IsLaundering* label was applied after simulation. Unlike PaySim, it contains no built-in

proxy of the target (e.g. no “oldBalance” fields). Any leakage would be due to model overfitting on patterns, not data design. We found no reports of inherent leakage.

- **IEEE-CIS:** The main risk is subtle data *split* leakage in Kaggle usage (e.g. knowing which time periods go to validation). With correct chronological splitting, feature leakage is minimal. There are no simulated artifacts to leak. The anonymized features do not obviously include post-fraud signals. Some Kaggle kernels have tried to infer the test set (e.g. matching user agents), but these exploit UI quirks, not the dataset itself.
- **BankSim:** Some features correlate strongly with fraud purely by simulation design. For example, “Category” has a strong negative correlation and “MerchantID” a positive one [39†L3-L6] . These can leak information: a model might simply use `category=="X"` to predict non-fraud. Although not classical target leakage, these artifacts mean the dataset is easier to classify than real data. Care is needed: one might remove the Category field to avoid this shortcut.
- **Kaggle Credit Card:** It has no engineered post-event fields (no balances, no “is_stolen” flags). The only potential leak is the highly imbalanced class: if one oversamples naively, or uses information from the test set distribution, one could inflate performance. But the dataset itself has no explicit leakage source. It is purely raw transactional data (anonymized).
- **FD Handbook Synthetic:** The dataset’s fraud patterns are rule-generated, which is a kind of leakage if not intended. For example, all transactions above a threshold (\$220) were marked fraud in scenario 1 [87†L43-L49] . A model will immediately latch onto `TX_AMOUNT>220` if present. Similarly, scenario 3 multiplies certain customers’ amounts by 5. These clear patterns mean a model doesn’t have to “discover” complex behavior – it’s a deterministic rule. In practice, this is a feature of the synthetic scenario (for teaching), but it must be treated carefully: if we consider these as “leaks,” then yes, models can easily exploit them. Other than these rules, the rest of the data is a straightforward time series with labels applied.

Evidence for leakage decisions: We cited Korolov 2022 Kaggle competition commentary for balance leakage [102†L1-L7] and [39] for BankSim feature correlations. Kaggle notes also warned about PaySim’s balance leak (e.g. the Kaggle page for Mobile Money Fraud: “balance-related variables may introduce data leakage” [102†L4-L7]). The Handbook’s documentation itself admits fixed rules, so models can “cheat” on them by thresholding [87†L67-L69] .

Behavioural Modelling Assessment

We assessed each dataset’s ability to support longitudinal fraud analytics (tracking entities over time):

- **Repeated Entities:** IBM AML, IEEE-CIS, BankSim, and the Handbook data *all* have repeat occurrences of customers/accounts. IEEE-CIS: multiple transactions share card fields (though not a single ID). BankSim: each CustomerID appears many times over 180 days. FD-Handbook: each CUSTOMER_ID and TERMINAL_ID repeats. Kaggle CC has none (no IDs).

- **Temporal Continuity:** IBM AML and FD-Handbook have real timestamps enabling sequence models. BankSim has a day index. IEEE-CIS has a relative time offset (TransactionDT). Kaggle CC has a time offset but, again, no ID to tie those times to specific customers.
- **Customer History & Relationships:** Only IBM AML and FD-Handbook allow rich histories and relationships. IBM AML's graph model means a given customer (account) has thousands of transactions across banks. FD-Handbook tracks each customer's 6 months of transactions. IEEE-CIS data can *in principle* link transactions by card1–6 as a pseudo-customer, but identity fields are anonymous and inconsistent. BankSim has single-hop links (customer→merchant), no deeper linkages. Kaggle CC has no history.
- **Behavioral Drift:** IBM AML's generative model can create changing behavior over long simulations. BankSim's customers have daily variation drawn from distributions. FD-Handbook can simulate changing patterns (e.g. some customers get “compromised” mid-stream). IEEE-CIS and Kaggle CC have too short a span to study drift meaningfully.
- **Fraud Chains & Patterns:** Only IBM AML is designed for multi-step laundering chains (fan-in/out, cycles) [95†L329-L337] . FD-Handbook does not simulate multi-step money flows, but it does simulate chained events (e.g. repeated frauds on one customer's account). IEEE-CIS and Kaggle CC contain only single-step labeled events.

Conclusion: Longitudinal modeling is genuinely possible with IBM AML and FD-Handbook. These datasets contain entity IDs and timelines to build features like “number of past transactions,” peer relationships, and network patterns. BankSim is limited (only one-step customer→merchant). IEEE-CIS allows only modest sequential features (features D and V change over time, but no semantic continuity). Kaggle CC offers none.

Operational Suitability Assessment

We evaluate whether each dataset supports a realistic fraud-analyst workflow:

- **Evidence Assembly:** Data should allow an analyst to see the context of a flagged transaction. IBM AML: the analyst sees source/dest accounts, but no human-readable info (no names, no merchant). Not ideal. IEEE-CIS & Kaggle: virtually no evidence (just numeric features). BankSim: minimal (see customer age/gender, merchant category). FD-Handbook: best (customer and terminal IDs let an analyst find usage patterns; synthetic scenario descriptions could be used to explain).
- **Explanation & Alerts:** The ideal dataset would include explainable features (customer profile, transaction reason, etc). None has direct textual fields. In practice: IBM AML could explain a fraud by referencing account networks (e.g. “this account paid itself via multiple hops”), but lacks human context. IEEE-CIS/Credit: explanations are purely statistical (“PCA comp #12 was high”). BankSim: could explain, “unusual merchant category for this customer.” FD-Handbook: could use thresholds (“transaction >220 in scenario 1”) which is trivial.

- **Recommendations & Investigation:** A human-in-the-loop system needs things like link analysis. IBM AML's graph is great for network analysis. FD-Handbook's table can be joined with customer summary stats to mimic an analyst's bank statement review. IEEE/CC have none of this.
- **Auditability & Review:** All datasets can be used to audit model performance (they have ground truth). However, for audit trails, lacking real-world IDs is problematic. The Handbook and BankSim data come with ground-truth rules (so audit is conceptual). In an actual deployment, IBM AML's known label origins make it auditable (the ground truth is perfect by design) but it's not real-data audit.

Summary: **FD-Handbook synthetic** most closely mimics a decision-support scenario (with customer/terminal IDs and explicit rules), followed by **IBM AML** (good for graph-based alerts, but limited narrative). The Kaggle datasets do not support real-world workflows at all – they're purely classification benchmarks. BankSim lies in between (simple KYC info but still mostly static).

Responsible AI Assessment

- **Representativeness & Bias:** None of these datasets were sampled to reflect a diverse population. IBM AML is synthetic statistical – it *could* be made demographically balanced, but we have no evidence of demographics (no gender/age). IEEE-CIS and Kaggle Credit contain unknown biases (they come from specific customer segments; e.g. the Kaggle credit data was European customers). BankSim uses an age distribution (18–98 uniform) but not tied to real census. The Handbook data places transactions uniformly. Potential bias (e.g. underrepresented fraud on certain demographics) is not documented in any.
- **Fairness:** We cannot assess fairness without protected attributes. BankSim includes gender, but nowhere is a protected-class fairness analysis possible. IEEE-CIS anonymized everything (no gender, race, etc), so fairness assessment isn't possible.
- **Privacy:** All datasets preserve privacy. The credit card data is anonymized PCA. IEEE-CIS replaced all IDs. IBM AML is fully synthetic (no real PII). FD-Handbook is synthetic with no real PII. Thus privacy risk is negligible for all.
- **Synthetic Data Limitations:** The synthetic sets (IBM AML, BankSim, FD-Handbook) may fail to capture some real-world complexities. For instance, IBM AML's virtual world may not perfectly match a particular country's banking rules. Fraudsters' adaptive behavior is hard to simulate; any static rule (like in FD-Handbook) is a simplification. Synthetic labels are perfect (no noise or disputed cases), unlike real investigations. Users should note that models trained on synthetic data may not generalize to real data without careful calibration.
- **Governance:** Using synthetic or benchmark data for a production system requires governance (data lineage, model limitations, etc). For real datasets (IEEE, Kaggle), licensing must be respected (especially Kaggle's non-commercial clauses). All of these are research-use – any model built on them should be validated on real operational data if possible.

Final Recommendation

Recommendation: *Replace PaySim in the modeling pipeline.* PaySim's known issues (target leakage via balances) make it unsuitable for production development. Instead, we recommend a **two-pronged**

strategy: continue to use PaySim only as a demonstrative example (to illustrate concepts), but **train and evaluate models primarily on better datasets**. Specifically, we suggest **IBM AML** as the core modeling dataset for money-laundering scenarios (it is the most realistic synthetic AML data [109†L181-L189]) and complement it with a real credit-card fraud dataset (such as IEEE-CIS) for retail fraud. This approach retains the existing architecture but substitutes PaySim’s training data.

Rationale: IBM AML offers a vast, labeled simulation of realistic laundering patterns [97†L168-L172] , directly addressing the shortcomings of PaySim. It avoids balance-based leakage and provides multi-bank transaction graphs. The IEEE-CIS (or Kaggle CC) data add real-world fraud examples for e-commerce card payments, which IBM AML lacks. By *combining* IBM AML with a credit-card dataset, we cover a broad spectrum of fraud without relying on PaySim’s faulty signals. All candidate datasets require careful feature engineering, but IBM AML’s documentation [97†L168-L172] and the FD-Handbook’s methodology [122†L215-L222] provide guidance. Overall, this strategy maximizes data fidelity and mitigates past pitfalls, yielding a more trustworthy fraud detection foundation.

References

- Altman et al., *Realistic Synthetic Financial Transactions for Anti-Money Laundering Models* (2023) [97†L168-L172] [109†L181-L189] .
 - Zhang et al., *Explainable ML for Fraud Detection* (2021) [45†L1-L4] [49†L496-L502] .
 - *IEEE-CIS Fraud Detection* (Kaggle 2019) dataset description [45†L1-L4] [49†L496-L502] .
 - Lopez-Rojas & Axelsson, *BankSim: A Bank Payments Simulator* (2016) [29†L0-L8] [39†L3-L6] .
 - Zou et al., *Credit Card Fraud Detection* (arXiv 2019) [66†L1-L4] .
 - Le Borgne & Bontempi, *Fraud Detection Handbook (Ch. 3)* (2021) [116†L83-L90] [122†L195-L202] .
 - IBM GitHub *AML-Data* repository [3†L240-L248] .
 - *Fraud Detection Handbook simulated-data* (GitHub) documentation [122†L199-L202] [122†L215-L222] .
 - *Fraud-detection-handbook.github.io* (Fraud Detection Handbook) [122†L193-L202] [122†L261-L264] .
 - Other Kaggle and online resources on financial fraud datasets.
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